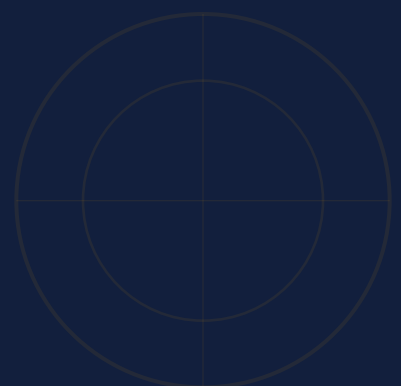

React-PDF + AI

The builder's guide to premium PDF generation.
Practical patterns for developers who ship.

LANDON MILES



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CHAPTER 01

Introduction

Why react-pdf and AI belong together – and what this book will teach you.

| What This Book Is

A practical guide for developers using AI coding agents – Claude Code, Cursor, Copilot – to generate PDFs with `@react-pdf/renderer`. You know React. What you may not know is how to structure a react-pdf project so AI produces premium output instead of generic templates. That gap is what this book closes.

Every page you're reading was built with these patterns. The theme file, component library, build scripts, and reference docs are the actual tools that produced this document – not hypothetical examples.

Built for AI-Assisted Development

The patterns in this book turn your AI coding agent from a generic template generator into a precision PDF builder.

| The Core Thesis

React-PDF is uniquely suited for AI-assisted development. It uses JSX – which AI models have deep training data on. It uses flexbox for layout – a familiar mental model. It renders server-side, so you can automate the entire pipeline.

However, the default output from an LLM rarely meets professional standards. Closing that gap is what the rest of this book is about – the specific architectural and design patterns that turn generic output into something premium.

Who This Is For

This book is for developers who want to move beyond "Hello World" PDFs. If you're building invoices, resumes, reports, or ebooks, and you want to use AI to speed up the process without sacrificing quality, this is for you.

We assume you know React and basic CSS. We don't assume you're an expert in PDF typography or the specific quirks of the react-pdf engine – we'll cover those as we go.

Why React-PDF?

While there are many ways to generate PDFs (Puppeteer, LaTeX, specialized APIs), @react-pdf/renderer offers the best balance of developer experience and AI compatibility. Because it's "just React," your existing skills – and your AI's knowledge of React patterns – apply directly.

Approach	Trade-off	AI fit
Puppeteer	Ships a headless browser; heavy and slow	Medium
LaTeX	Beautiful output, steep and arcane syntax	Low
PDF APIs	Fast to start, opaque and hard to customize	Low
React-PDF	Component model you already think in	High

Why AI fit matters

Your AI agent has seen millions of React components. When the layout is JSX, it can reason about structure, reuse your design tokens, and edit a page the same way it edits any component – no PDF-specific dialect to learn first.

CHAPTER 02

React-PDF Fundamentals

Components, styling, layout, and the rules of the rendering engine.

Component Hierarchy

Every react-pdf document has a required root and page structure. Inside a Page, supported react-pdf primitives can be direct children. Invalid trees may warn, omit content, or fail to render.

```
tsx
import { Document, Page, View, Text, Image }
from '@react-pdf/renderer';

const MyDoc = () => (
  <Document>
    <Page size="LETTER">
      <View>
        <Text>Ordinary visible copy belongs in Text.</Text>
        <Image src="photo.png" />
      </View>
    </Page>
  </Document>
);
```

The rules:

- Document is the root – its rendered direct children must be Page elements
- Page defines a physical page – size, orientation, margins
- View is an optional container (like div) – it handles layout via flexbox
- Text renders ordinary visible copy – Page can contain Text directly
- Image renders PNG and JPG – no SVG files (use Svg components instead)
- No DOM elements inside Document – custom components must resolve to react-pdf primitives

⚠ Common Mistake

A raw string directly under Page or View triggers a console warning and is omitted. Put ordinary copy in Text; text-capable primitives such as Link are exceptions.

Page Sizes

Set the physical page with the `size` prop on `Page`. Pass a named size, or a custom `[width, height]` in points (72pt = 1 inch); orientation flips it to landscape.

```
tsx
// Named sizes
<Page size="LETTER">    { /* 612 x 792 pt */ }
<Page size="A4">        { /* 595 x 842 pt */ }
<Page size="LEGAL">     { /* 612 x 1008 pt */ }

// Custom trim: [width, height] in points
<Page size={[468, 648]}> { /* 6.5in x 9in book */ }

// Landscape
<Page size="LETTER" orientation="landscape">
```

Named sizes include A0–A10, B0–B10, C0–C10, LETTER, LEGAL, TABLOID, EXECUTIVE, and FOLIO. For ebooks and reports, US Letter (612x792) and A4 (595x842) handle almost everything – pick one and use it for every page.

Keep it uniform

Mixing page sizes in one document reads as a bug. This book's build runs `pdfinfo` and fails if any page isn't a uniform LETTER, so an accidental A4 or landscape page can't slip through.

Images

Image renders PNG, JPG, and base64 data URIs – but not SVG files (use the `Svg` component for those). The one real gotcha: without explicit dimensions an image renders at its native pixel size, where 1 pixel = 1 point. A 1000x500 image becomes 1000x500pt – about 14 x 7 inches – and overflows the page.

```
tsx
// Always size it - width/height, or a width plus objectFit
<Image src="/assets/photo.jpg"
style={{ width: '100%', height: 200, objectFit: 'cover' }} />

// Authenticated URLs: pass a source object with headers
<Image source={{
  uri: 'https://api.example.com/image/123',
  method: 'GET',
  headers: { Authorization: 'Bearer <token>' },
}} />
```

⚠ Size every image

An unsized image is the most common first-render overflow. Set width and height (or a width plus `objectFit`) on every `Image`, and prefer PNG for anything that needs transparency.

Styling System

React-PDF styles are JavaScript objects, like React Native – not CSS strings or className props. Tokenize reusable choices; name one-off geometry locally.

```
tsx
import { StyleSheet } from '@react-pdf/renderer';
import { colors, spacing, borders } from '../../styles/theme';
const styles = StyleSheet.create({
  container: {
    flexDirection: 'row',
    padding: spacing.lg,
    backgroundColor: colors.neutral[50],
    borderRadius: borders.radius.md,
  },
});
```

Supported Layout: Flexbox First

Column is the default. Normal flow uses flexbox – no Grid, floats, or inline/block display. Relative and absolute positioning, plus inline style objects, are valid.

Supported	Not Supported
flexDirection, alignItems, justifyContent	CSS Grid (grid-template-*)
width, height, minWidth, maxWidth	float, clear
margin, padding (all sides)	display: inline / block / table
position: relative / absolute	box-shadow, text-shadow
border (width, color, style, radius)	CSS animations / transitions
backgroundColor, opacity, @media queries	calc(), CSS variables, background-image

Units

The default unit is points (pt). 1 inch = 72 points. You can also use strings with unit suffixes.

Unit	Example	Notes
Points	16	Default – no suffix needed
Inches	"1in"	1in = 72pt
Millimeters	"25.4mm"	25.4mm = 1in = 72pt
Centimeters	"2.54cm"	2.54cm = 1in = 72pt
Percentage	"50%"	Relative to parent container

Colors

Hex, RGB, RGBA, HSL, and named colors all work. Define your palette once in a theme file and reference the constants everywhere.

```
tsx
export const colors = {
  navy: '#0f2942', // hex
  gold: 'rgb(245, 184, 73)', // rgb
  muted: 'hsl(215, 16%, 47%)', // hsl
  overlay: 'rgba(15, 41, 66, 0.6)', // rgba with alpha
};

// Reference everywhere – change once, update the whole document
<Text style={{ color: colors.navy }}>Heading</Text>
```

Font Registration

Three font families are built in: Courier, Helvetica, and Times-Roman, each with bold and italic variants. For anything else, you register fonts explicitly.

```
tsx
import { Font } from '@react-pdf/renderer';
import path from 'path';
const dir = path.resolve(__dirname, '../fonts');
Font.register({ family: 'Inter', fonts: [
  { src: path.join(dir, 'Inter-Regular.ttf'), fontWeight: 400 },
  { src: path.join(dir, 'Inter-SemiBold.ttf'), fontWeight: 600 },
  { src: path.join(dir, 'Inter-Bold.ttf'), fontWeight: 700 },
] }); // + Medium 500 and italics in the real project
```

⚠ Font Limitations

Only TTF and WOFF formats work – OTF files and variable fonts are not supported. Missing styles fail; missing weights resolve to the nearest registered weight.

Using Registered Fonts

Reference a registered family by name and request a weight or style. Register the variants you intend to use so nearest-weight fallback never changes the design unexpectedly.

Style	Resolves to	Common use
fontWeight: 400	Inter-Regular.ttf	Body copy, captions
fontWeight: 600	Inter-SemiBold.ttf	Headings, bold text
fontWeight: 700	Inter-Bold.ttf	Heaviest emphasis

Page Breaking

- `wrap={true}` on `Page` – allows content to overflow to next page (default)
- `break={true}` on `View/Text` – forces a break when the ancestor `Page` wraps
- `fixed={true}` on `View` – repeats on each subpage produced by its `Page`
- `minPresenceAhead={number}` – moves forward when too little can follow on a wrapping `Page`
- `orphans/widows` on `Text` – minimum lines around a split on wrapping pages

⚡ Tip

Use `Text`'s `render` prop for `pageNumber` and `totalPages`. `View`'s typed `render` prop exposes `pageNumber` and `subPageNumber`. Example: `render={({ pageNumber }) => pageNumber}`

A Complete Example

A fixed footer, a forced break before a new chapter, and a card that refuses to split – combined in one tree:

```
tsx
<Page wrap>
  <View fixed style={footer}>
    <Text render={({ pageNumber, totalPages }) =>
      `${pageNumber} / ${totalPages}` />
    </View>
    <View break>
      <Text>Chapter 2 starts on a fresh page</Text>
    </View>
    <View wrap={false}>
      {/* This card stays whole or moves down */}
    </View>
  </Page>
```

Putting It Together

These fundamentals – the document/page structure, page sizes, images, JavaScript-object styling, flexbox layout, font registration, and page breaking – are the building blocks. Every page in this book uses them.

Internalizing these constraints is the first step toward mastery. Once you stop reaching for web-specific CSS and embrace the react-pdf primitives – no grid, no CSS display: inline/block, no CSS variables – you can leverage the engine's predictable layout behavior to build complex, multi-page documents with confidence.

The next chapter shows how to organize these building blocks into a project architecture that AI agents can work with efficiently. The goal: a structure where AI can edit one page without reading (or breaking) the rest.

Concept	Key Rule
Component structure	Document > Page; Page accepts react-pdf primitives directly
Page sizes	size prop – named sizes or [width, height] in points; keep uniform
Images	Always set dimensions – unsized renders at 1px = 1pt and overflows
Styling	JavaScript objects or arrays – no CSS text or className
Layout	Flexbox for normal flow – default direction is column
Fonts	Register TTF/WOFF explicitly – no synthesized bold/italic
Page breaks	wrap, break, fixed, minPresenceAhead props
Units	Points (default), inches, mm, cm, percentages

CHAPTER 03

Project Architecture for AI Agents

Structure your codebase so AI can edit one page without breaking the rest.

Why Monoliths Break for AI

The most common react-pdf mistake is putting everything in one file. Your entire 30-page document lives in a single 2,000-line component. You paste it into an AI prompt. The AI edits page 12 and accidentally breaks the styling on page 4.

Big files also strain AI attention – Chapter 5 covers the token math in detail. The structural fix in this chapter is what stops the cross-page breakage even before tokens enter the picture.

File-Per-Page Pattern

Split every page into its own component file. One file, one page, one concern.

```
bash
src/
pages/
01-cover/01-cover.tsx
02-toc/01-toc.tsx
03-introduction/00-title.tsx      # chapter divider
03-introduction/01-introduction.tsx
components/  # Header, Footer, ChapterTitle, CodeBlock, Table, ...
styles/
theme.ts      # design tokens
shared.ts     # shared StyleSheet
manifest.ts   # chapter structure (source of truth)
Document.tsx # maps the generated registry into one <Document>
build.tsx     # two-pass render to output/react-pdf-ai-builders-guide.pdf
```

Why This Works

- Page files stay small and focused – often a few dozen lines
- AI can edit one page (e.g. 05-architecture/07-naming-conventions.tsx) without touching the others
- You give AI context: the page file + theme.ts + relevant components = ~3,000 tokens
- Components enforce consistency – every content page uses the same Header and Footer

Assembly File

The generated registry imports every page. Document.tsx imports that ordered list and maps it into one Document, so the assembly never drifts from the page tree.

```
tsx
// Document.tsx (metadata abridged)
import React from 'react';
import { Document } from '@react-pdf/renderer';
import { allPages } from './registry';
const EbookDocument: React.FC = () => (
  <Document
    title="React-PDF + AI: The Builder's Guide"
    author="Landon Miles"
  >
    {allPages.map((Page, index) => <Page key={index} />)}
  </Document>
);
export default EbookDocument;
```

Build Script

The simplest build script renders the document to a PDF in one call – run it with tsx or ts-node. (This book's own build.tsx adds a second render pass to fill in real TOC page numbers.)

```
tsx
// build.tsx
import ReactPDF from '@react-pdf/renderer';
import EbookDocument from './Document';
ReactPDF.render(<EbookDocument />, './output/react-pdf-ai-builders-guide.pdf')
  .then(() => console.log('PDF generated: output/react-pdf-ai-builders-guide.pdf'));
```

Clickable Navigation

react-pdf turns a Link whose src is #id into an internal jump to the element that carries that id – usually a Page. Give one helper ownership of the destination name so the link and its target can never drift apart; the TOC in this book is built exactly this way.

```
tsx
// manifest.ts - one helper owns the destination name
export const chapterDestinationId = (num) => `chapter-${num}`;

// Destination: the chapter divider page carries the id
<Page id={chapterDestinationId('03')}> ... </Page>

// Source: the TOC row links to that same id
<Link src={`#${chapterDestinationId('03')}`}>
<Text>Project Architecture</Text>
</Link>
```

Real Page Numbers

The page a chapter lands on isn't known until layout runs, so a static TOC can't hardcode it. This book renders twice: pass 1 renders the PDF, pdftotext finds the page holding each "CHAPTER NN" marker and writes the positions to toc-positions.json, then pass 2 re-renders with the numbers filled in. The TOC reserves its number column on both passes, so layout is identical and the pass 1 positions stay valid.

Keep the marker unambiguous

The extractor matches a standalone uppercase "CHAPTER NN" line. Don't print that exact string in body text, or two chapters could resolve to the wrong page – the build fails loudly if a marker appears on more than one page.

PDF Metadata

The Document component accepts metadata props that get embedded in the PDF's info dictionary. Viewers show them in tab and title bars, search tools and file browsers index them, and assistive technology reads them – so setting them is a few props of effort for a more professional, accessible deliverable.

Prop	What it sets
title	Shown in viewer tabs and title bars; used when shared or indexed
author	Person or organization credited as the author
subject	One-line description of the document's topic
keywords	Comma-separated terms that search tools index
creator	The application the content was authored in
producer	The library that generated the PDF (react-pdf sets a default)

Set title and author at minimum

They surface in viewer chrome and file listings and are read by screen readers, so a document without them looks unfinished when shared. Drive the values from your data so every generated file is labeled correctly.

Shared Components

Components that appear on multiple pages live in the components/ folder. This is where consistency comes from.

Component	Purpose	Used By
Header	Book title + section name at top	Every content page
Footer	Page number + branding at bottom	Content pages + TOC
ChapterTitle	Full-page chapter opener	Start of each chapter
ContentPage	Standard page with header/footer	All content pages
CodeBlock	Styled code display	Technical chapters
TipBox / WarningBox	Callout boxes	Any page with key info
Table	Flexbox-based data table	Comparison/reference pages
BulletList	Consistent bullet lists	Throughout
Icons (SVG)	Vector icons for visual interest	Headers, callouts, lists

Naming Conventions

Chapter files follow a strict pattern: NN-chapter/, with 00-title.tsx for the divider and NN-topic.tsx for content. Cover, TOC, and conclusion are chrome exceptions: each has one 01- file and no divider. Components are PascalCase.tsx. Numeric prefixes keep files sorted and make the target unambiguous. The chapter structure itself lives in src/manifest.ts.

⚠ Anti-Patterns

Avoid these structures – they make AI editing harder and more error-prone: monolithic single-file documents, inline styles on every element (use StyleSheet.create), deeply nested component trees that span pages, importing all components everywhere instead of just what's needed.

The Pattern at a Glance

Chapter paths resolve to a handful of shapes; chrome uses the exception above. Once your AI internalizes the rule, it can name new files correctly – and you can predict where anything lives.

Pattern	Example	What it is
NN-chapter/	05-architecture/	One folder per chapter
00-title.tsx	00-title.tsx	The chapter divider page
NN-topic.tsx	07-naming-conventions.tsx	A single content page
PascalCase.tsx	CodeBlock.tsx	A reusable component
manifest.ts	src/manifest.ts	The chapter structure

How Changes Propagate

When you change a design token in theme.ts, every consumer updates on the next build. Change primary[800], and the heading roles that use it update. Change typography.body.fontSize, and copy using the body preset adjusts. This is centralized decision-making with distributed rendering.

The same applies to shared components. Update the Footer component, and every content page gets the new footer. Fix a bug in CodeBlock, and every code example benefits. You never have to hunt through a monolith file to find every instance of a repeated pattern.

A useful rule of thumb: if a style block or component appears on 3+ pages, extract it into the shared folder. If it only appears once, keep it local to that page file. This keeps the shared layer focused and avoids bloat while still centralizing everything that needs to be consistent.

One Edit, Many Pages

The system hinges on importing values instead of repeating them. Each heading role references a named color, so a token edit reaches every component that consumes it on the next build:

```
ts
// src/styles/theme.ts - reusable design tokens (excerpt)
export const colors = {
  primary: { 800: '#121F3D' /* ...9 more steps */ }, // change here ...
  accent: { 500: '#F0A000' /* ...9 more steps */ }, // ...every consumer follows
};
```

You change...	What updates	Blast radius
A design token	Every component consuming it	Targeted to book-wide
A shared component	Every page using it	Many pages
One page file	Only that page	Single page

CHAPTER 04

Specifying a Design Language

Define your visual system once. Use it everywhere. Let AI enforce it.

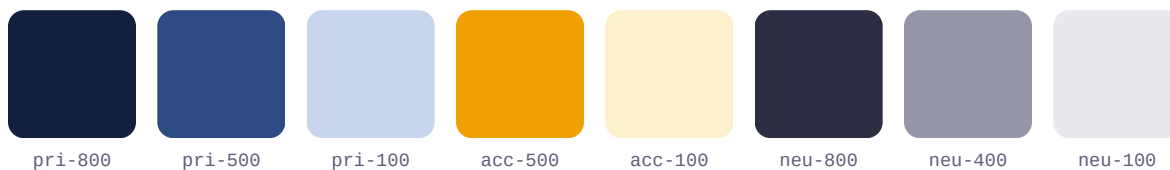
Why AI Needs Explicit Design Tokens

Without a design system, every AI-generated page is a coin flip. The AI picks `fontSize 14` on one page and `16` on the next. It uses `#333` for body text here and `#444` there. The result looks like five different people designed it.

A design token file solves this. It centralizes reusable visual decisions: colors, font sizes, spacing, and border radii. Include it in every AI prompt so generated pages inherit the same constraints.

Color Palette

Pick one primary role, one accent role, and a neutral scale. Three to five core roles handle most document design; tonal steps provide useful variation without adding new roles.



This book uses navy primary (`#121F3D`) with amber/gold accent (`#F0A000`) and slate neutrals. The navy conveys trust and professionalism. The gold adds warmth and draws the eye to important elements.

⚡ Tip

Use darker primary shades (700–900): 800–900 for major headings and surfaces, 700 for minor headings. Use the accent for highlights and chapter numbers, and semantic colors for status callouts. Neutrals handle body text, dividers, and secondary information.

Tested Color Palettes

Generate your own at colors.co, or start from one of these. Each pairs a dark primary with a saturated accent; keep enough contrast between the two that they read as distinct side by side – a navy primary with a near-navy accent looks muddy. A muted mid-tone secondary fills borders and captions.

Scheme	Primary	Accent	Best for
Navy + Gold	#1a2b4a	#d4a843	Financial, legal, executive
Teal + Amber	#0f766e	#f59e0b	Tech & product docs
Charcoal + Blue	#1e293b	#3b82f6	SaaS & analytics
Black + Red	#18181b	#dc2626	Editorial, bold statements
Warm Earth	#44403c	#b45309	Nonprofit, sustainability

⚡ Wire the palette into tokens

Drop the primary and accent hex values into `colors` in `theme.ts` once, and derive the secondary as a desaturated mid-tone of your primary. Every component reads from there, so re-skinning the whole document is a small change – and your AI agent picks up the new palette automatically.

Typography Scale

Define every text size upfront. No magic numbers in page components. Page chrome – this book's 42pt cover title, 32pt chapter titles – comes from a separate `fontScale` token.

<code>display</code>	36pt	Hero/display text
<code>h1</code>	26pt	Top-level headings
<code>h2</code>	20pt	Section headers
<code>h3</code>	16pt	Subsections
<code>h4</code>	13pt	Labels, small headers
<code>body</code>	11pt	Main reading text
<code>bodySmall</code>	9.5pt	Secondary text
<code>caption</code>	8.5pt	Footnotes, captions
<code>code</code>	9pt	Code blocks
<code>codeSmall</code>	8pt	Dense code, labels

One Source of Truth

Encode the scale as an object in `theme.ts` and read every size from it. When you tell your AI agent to build a page, it pulls from these tokens – so headings stay proportional and nothing drifts into a stray 14.5pt.

```
tsx
export const typography = {
  display: { fontSize: 36, fontFamily: fonts.heading, fontWeight: fontWeight.bold },
  h2:      { fontSize: 20, fontFamily: fonts.heading, fontWeight: fontWeight.semibold },
  body:    { fontSize: 11, fontFamily: fonts.body,   fontWeight: fontWeight.regular },
} as const;
```

Spacing Scale

Use a mostly 4-point grid with named 1pt and 2pt micro exceptions. Reusable gaps and padding come from it; page geometry has separate named tokens:

```
typescript
export const spacing = {
  none: 0, micro: 1, xxs: 2,      // zero + micro exceptions
  xs: 4, sm: 8, md: 12, lg: 16,   // 4pt core
  xl: 24, xxl: 32, xxxl: 48,     // major spacing
};
```

Theme File

Tokens live in theme.ts. Pages consume them directly or through shared styles/components; reusable design values are not hardcoded.

```
typescript
import { colors, fonts, typography, spacing } from '../styles/theme';
const s = StyleSheet.create({
  title: {
    ...typography.h1,
    color: colors.primary[800],
    marginBottom: spacing.lg,
  },
});
```

The AI Context Pattern

When prompting AI to create a new page, include theme.ts in the context. The AI will use your defined tokens instead of inventing its own values.

Border and Radius Tokens

Define border widths and corner radii alongside your other tokens. Without these, AI mixes 4pt rounded corners on one card with 8pt on another – subtle but noticeable.

```
typescript
export const borders = {
  thin: 0.5,    // Dividers, table cell borders
  medium: 1,    // Card borders, table header rules
  thick: 2,     // Callout left borders, emphasis
  radius: {
    xs: 1.5,    // Fine details, compact elements
    sm: 3,      // Callout boxes, badges, tags
    md: 6,      // Cards, code blocks, tables
    lg: 10,     // Feature cards, hero elements
  },
};
```

Semantic Color Tokens

Beyond your core palette, define semantic colors for meaning: success (green), warning (amber), error (red), info (blue). Include light variants for callout backgrounds.

```
typescript
success: '#2D8B4E', successLight: '#F0F9F4',
warning: '#D98E00', warningLight: '#FEF8E6',
error: '#C43333', errorLight: '#FEF3F3',
info: '#2E6BB5', infoLight: '#EDF1F8',
```

⚡ The Complete Token Set

Start with four core categories: colors (5-8 base values), typography (7-9 sizes), spacing (7-9 values), and borders (3 widths + 4 radii), plus semantic colors for callouts. Then promote recurring magic numbers into new tokens – this book's theme.ts grew to 15 categories.

Design Token Checklist

Category	Count	Example
Colors (primary + accent)	20 shades (10 each)	#121F3D navy, #F0A000 gold
Neutrals + semantic	5-10 + 4 pairs	Gray scale + success/warning/error/info
Typography	10 presets	8pt codeSmall to 36pt display
Spacing	9 positive + zero	1pt micro to 48pt xxxl
Borders	3 widths + 4 radii	0.5pt thin to 10pt lg radius

Auditing Your Tokens

Run your theme.ts through this starter list before you build a single page – then let it grow. This book's theme reached 15 categories, from font weights to icon sizes:

- Reusable design values live in theme.ts; unique geometry uses named local constants instead of scattered literals.
- Each scale has an intentional progression – a mostly 4pt spacing grid with micro exceptions and a curated type hierarchy.
- Semantic colors map to meaning – success, warning, error, info – not to a specific shade you might want to swap later.
- Names describe role, not appearance – primary[800] over navyDark, so a rebrand is a centralized palette change.

CHAPTER 05

Tokenization and Context Windows

How LLMs read your code – and how to make every token count.

What Tokens Are

LLMs don't read characters or words. They read tokens – subword units that the model was trained to recognize. In English text, one token is roughly 4 characters or 0.75 words. Code tokenizes differently because of syntax characters.

This matters for react-pdf because your page components are code. The angle brackets, prop names, style objects, and JSX structure all consume tokens. A simple page component with a few Text elements and a StyleSheet might be 200-400 tokens. A complex page with tables and code blocks could be 600-1,000 tokens.

Context Window Math

Every AI model has a context window – the total amount of text it can process in a single conversation turn. Specific models change frequently, but the categories remain stable:

Category	Context Window	Effective Working Memory
Standard models	32K-128K tokens	~4,000-12,000 tokens (sweet spot)
Large-context models	128K-200K+ tokens	~10,000-20,000 tokens (sweet spot)
Extended-context models	500K-1M+ tokens	~20,000-40,000 tokens (sweet spot)

The 'Lost in the Middle' Problem

Research shows LLMs pay most attention to the beginning and end of their context. Content in the middle gets less attention. A 15,000-token monolithic PDF file means your AI is likely ignoring the pages in the middle – exactly where bugs hide.

Token Cost of React-PDF Code

JSX is token-expensive compared to plain text. Here's what typical react-pdf structures cost:

Structure	Approximate Tokens
A StyleSheet.create() with 10 style objects	200-350
A simple page (title, 3 paragraphs)	150-250
A complex page (table, code block, callouts)	400-800
A shared component (Header or Footer)	400-500
A theme.ts with full design tokens	1,700-1,900
A 30-page monolith file	8,000-15,000

Practical Strategy

When you ask AI to edit a page, it needs context. Here's the optimal context budget:

- theme.ts (design tokens): ~1,800 tokens
- Relevant shared components: ~300-500 tokens
- The page file being edited: ~200-600 tokens
- Your instructions: ~100-300 tokens
- Total: 2,400-3,200 tokens of context – well within any model's sweet spot

Compare that to the monolith approach: 8,000-15,000 tokens dumped into context, with your edit instructions buried somewhere inside. The file-per-page architecture isn't just about code organization. It's about giving AI the right amount of focused context.

Reducing Token Waste

- Extract repeated styles into shared.ts – define once, reference everywhere
- Use short but descriptive variable names – "s" for local styles is fine
- Don't add comments that restate the obvious – "// Render the title" above a title render
- Keep page components focused – if a section could be its own page, make it one
- Use docs/ for concise project guidance; keep long-form research in reference/

Prompt Sizing in Practice

Here's what an efficient AI prompt looks like. Notice how small and focused the context is:

text

Context files (paste these):

1. theme.ts (your design tokens)
2. ContentPage.tsx (the wrapper component)
3. 07-tokenization/04-reducing-waste.tsx (target page)

Instruction:

"Add a new section called 'Prompt Sizing' after the BulletList. Use a SectionHeading for the heading and styles.body for the paragraph. Include a CodeBlock showing an example prompt."

Total context: roughly 2,800 tokens. The AI has everything it needs to produce code that matches your design system, uses the correct component wrapper, and fits the existing page structure. No guesswork required.

Project Docs vs. Research

Keep concise project guidance – design tokens, component API, conventions – in docs/. Reserve reference/ for long-form research. Point AI to the smallest relevant docs instead of loading all source files.

Token Budget Template

For each AI edit session, aim for this budget: theme (1,800) + components (350) + target page (450) + instructions (200) = 2,800 tokens. In a 128K context window, that leaves about 98% available for reasoning and output.

Measure Your Own Files

Estimates lie. Real numbers don't. Run this in your project root for a per-file token estimate (4 chars $\approx$ 1 token):

```
bash
# Approximate token count per .tsx file, largest first
find src -name '*.tsx' -exec wc -c {} \; \
| awk '{ printf "%6d tokens  %s\n", $1/4, $2 }' \
| sort -rn | head -20
```

Page or feature files over 1,500 tokens are worth reviewing for a split. Cohesive source-of-truth files such as theme.ts can stay larger when dividing them would make the system harder to follow. Then sum the files an AI actually needs for one edit (theme + components + target page + instructions). A total under 4,000 is compact even for a 32K model and uses just over 3% of a 128K context window.

CHAPTER 06

Avoiding AI Slop

What makes output look cheap – and the specific fixes that make it premium.

| What AI Slop Looks Like

You've seen it. The AI generates a PDF and it looks... generic. Default font. Random spacing. Every page feels like a different document. Here's what specifically goes wrong:

- Default Helvetica everywhere – no typographic personality
- Inconsistent font sizes – 14pt here, 16pt there, no clear hierarchy
- Cramped margins – content pushed to the edges with no breathing room
- Random colors – #333, #666, #007bff borrowed from different templates
- No visual hierarchy – body text and headings nearly the same size
- Walls of text – no callout boxes, no bullet lists, no visual breaks
- Lorem ipsum remnants – placeholder text the AI forgot to replace

| Root Causes

AI slop isn't the AI's fault. It's a prompt and structure problem.

- Lazy prompts: "Make me a PDF" gives the AI nothing to work with
- No design system: without tokens, the AI improvises – badly
- No visual QA: you accept the first output without reviewing the render
- Monolith structure: the AI can't focus on one page at a time
- No iteration: premium output takes 2-3 passes, not one

⚠ The First Draft Rule

AI's first attempt is a rough draft. Always. It gives you the structure and general direction, but the spacing, sizing, and visual balance need refinement. If you ship the first draft, you ship slop.

Slop vs. Premium – Side by Side

Heading Styles

✗ Slop

fontSize: 16, centered, default font, no color, no spacing above or below. Blends into body text.

✓ Premium

Uses typography.h2 (20pt), primary color, 16pt marginBottom, 24pt marginTop. Clear hierarchy break.

Page Margins

✗ Slop

padding: 20 on all sides. Content cramped against edges. Feels claustrophobic.

✓ Premium

54pt horizontal, 60pt vertical margins. Content has room to breathe. Professional whitespace.

Color Palette

✗ Slop

Pure black (#000) ink with a saturated default-blue link. Harsh contrast, screen-not-print feel.

✓ Premium

Deep navy ink, never pure black, with one restrained gold accent – a few tokens, used consistently everywhere.

⚡ The Pattern

Slop comes from accepting library defaults; premium comes from a small set of deliberate tokens applied without exception. The fix is rarely more design – it is fewer decisions, made once, and reused on every page.

Iteration Workflow

Premium output requires a loop, not a single prompt:

- Pass 1: Generate structure and content – get the text right
- Pass 2: Apply design system – enforce tokens, fix spacing
- Pass 3: Visual QA – render to PNG, review, fix alignment and balance
- Optional Pass 4: Polish – add callouts, refine typography, check page breaks

⚡ The Prompt That Fixes Slop

Instead of "make me a page about X", try: "Create a content page using `ContentPage` wrapper. Use typography from `theme.ts`: `h2` for section titles, `body` for paragraphs. Include a `TipBox` callout and a `BulletList`. Match the spacing and style of `05-architecture/01-architecture.tsx`."

Knowing When to Stop

The loop has a floor. Past a certain point, more passes add churn, not quality – and over-polishing reintroduces its own kind of slop. Stop when:

- The page renders cleanly with no orphaned headings, split callouts, or stray bullet dots
- Every reusable color, font, and spacing value traces back to a named token
- The layout reads at a glance – clear hierarchy, comfortable density, balanced whitespace
- A fresh read surfaces only nitpicks you would not pay to fix

⚠ Diminishing Returns

If your AI keeps "improving" a page that already passes QA, you are spending tokens to move pixels. Lock the page, commit it, and move the loop to the next one.

Prompt Templates That Prevent Slop

The prompts that produce premium react-pdf output share one trait: they front-load constraints so the AI can't improvise. The template below is copy-paste ready – fill the bracketed slots and run it. Six fuller templates ship in the repo's templates/ directory.

New Page

```
Build a new ContentPage for [TOPIC].
```

```
Reference: src/pages/[EXISTING_PAGE].tsx for structure.
```

```
Design system: src/styles/theme.ts (tokenize reusable choices).
```

```
Components: SectionHeading, BulletList, CodeBlock, TipBox, Table.
```

```
Requirements:
```

- Use ContentPage wrapper with sectionTitle="[SECTION]"
- Import styles from '../..../styles/shared'
- Every text run must resolve to a registered family and weight
- Use shared typography styles; pair fontFamily + fontWeight when defining either
- wrap={false} on elements that must not split across pages
- No Helvetica, no emoji, no hardcoded colors

⚡ The Negative Constraint Pattern

Three or four negative constraints ("no Helvetica," "no emoji," "no hardcoded colors") consistently produce cleaner output than twice as many positive instructions. LLMs predict the most likely next token – negative constraints override bad defaults and close escape hatches the AI would otherwise use.

Copy-Paste Test

Open a fresh AI session with no prior context. Paste only the template plus the files it references. If the generated page matches your existing pages, the prompt is good enough; if not, the gap reveals which constraint is missing.

CHAPTER 07

Design Challenges and Solutions

What works, what breaks, and the workarounds that actually hold up.

| What Works Well

React-PDF will frustrate you if you fight it. Don't. These patterns work cleanly – lean into them and you'll build faster than you expect:

- Single-column layouts with clear heading hierarchy
- Two-column layouts using `flexDirection: "row"` with fixed widths
- Full-width colored banners and pull quotes with accent borders
- Consistent card patterns: bordered View with padding and `borderRadius`
- SVG decorative elements (lines, circles, icons)
- Bullet lists and tables built with flexbox rows
- Page wrapper inserts headers/footers; `fixed={true}` repeats them on wrapped pages

| What to Avoid

The flip side: a handful of patterns that feel reasonable in a browser but quietly break in PDF. Reach for them and you'll spend your afternoon debugging layout instead of shipping pages:

- Percentage heights in an unsized parent – they resolve against nothing and collapse
- Floated or grid layouts – use flexbox rows and columns for normal flow
- `position: "fixed"` – no such style value; repetition comes from the `fixed={true}` prop
- Deeply nested Views chasing pixel-perfect alignment – flatten and let flex do the work

⚠ Watch out

When a layout misbehaves, an unsupported CSS property is a common cause. Check the supported-style list before you start bisecting, then inspect wrapping, content height, and flex constraints.

Charts and Data Viz

react-pdf has no charting primitive, and browser charting libraries (Chart.js, Recharts, D3's DOM rendering) don't run in the renderer – though D3's pure math modules (d3-scale, d3-shape) can compute path data for the Svg primitives. Two routes cover almost every case: render a complex chart externally to a PNG and place it with Image, or build a simple one from Views.

For anything intricate – area, scatter, stacked – render it in a headless browser or node-canvas, export a PNG at 3–4x the placed size, and drop it in with a caption. For bars, progress, and gauges, nest two Views and drive the inner width off a percentage:

```
tsx
const Bar = ({ label, value, max }) => (
  <View style={{ flexDirection: 'row', alignItems: 'center', marginBottom: 8 }}>
    <Text style={{ fontSize: 9, width: 80 }}>{label}</Text>
    <View style={{ flex: 1, height: 12, backgroundColor: '#e5e7eb', borderRadius: 2 }}>
      <View style={{ width: `${(value / max) * 100}%`, height: 12,
        backgroundColor: '#3b82f6', borderRadius: 2 }} />
    </View>
  </View>
);
```

⚡ Match exported charts to your tokens

When you render a PNG chart externally, feed it the same hex values from theme.ts so it doesn't clash with the document, and export at 3–4x the placed size so it prints crisp – a 2x export only reaches roughly 150 DPI.

What Doesn't Work (and Fixes)

✗ CSS Grid Layouts

Not supported. Use nested flexbox with explicit widths instead:

```
tsx
// Three-column "grid" using flexbox
<View style={{ flexDirection: 'row' }}>
  [{ 'Column 1', 'Column 2', 'Column 3' }.map((label) => (
    <View key={label} style={{ width: '33%', padding: spacing.sm }}>
      <Text>{label}</Text>
    </View>
  ))}
</View>
```

✗ Drop Shadows

box-shadow is not supported. Use a subtle border or an offset nested View:

```
tsx
// Fake shadow via darker bottom/right borders
<View style={{
  borderWidth: borders.medium,
  borderColor: colors.neutral[200],
  borderRadius: borders.radius.md,
  borderBottomWidth: borders.thick,
  borderRightWidth: borders.thick,
  borderBottomColor: colors.neutral[300],
  borderRightColor: colors.neutral[300],
  padding: spacing.lg,
  backgroundColor: colors.white,
}}>
  <Text>Card with faux shadow</Text>
</View>
```

✗ Text Wrapping Around Images

Float is unsupported. Use a row with named one-off geometry:

```
tsx
const mediaSize = { width: 120, height: 90 };
<View style={{ flexDirection: 'row', gap: spacing.lg }}>
  <Image src="photo.png" style={mediaSize} />
  <View style={{ flex: 1 }}>
    <Text>Text flows in a column beside
    the image, not wrapped around it.
  </Text>
</View>
</View>
```

✗ Gradient Backgrounds

CSS gradients need SVG. Name the layer height once:

```
tsx
const GRADIENT_HEIGHT = 100;
<View style={{ position: 'relative', height: GRADIENT_HEIGHT }}>
  <Svg style={{ position: 'absolute', top: 0, left: 0, right: 0, bottom: 0 }}
    viewBox={`0 0 ${page.contentWidth} ${GRADIENT_HEIGHT}`}>
    <Defs>
      <LinearGradient id="bg" x1="0" y1="0"
        x2="1" y2="0">
        <Stop offset="0%" stopColor={colors.primary[700]} />
        <Stop offset="100%" stopColor={colors.primary[500]} />
      </LinearGradient>
    </Defs>
    <Rect width={page.contentWidth} height={GRADIENT_HEIGHT}
      fill="url(#bg)" />
  </Svg>
  <Text style={{ color: colors.white, padding: spacing.lg }}>
    Content over gradient
  </Text>
</View>
```

Recipe: Professional Table

Tables in react-pdf are built with flexbox rows, not HTML table elements. The key ingredients: a dark header row with white semibold text, alternating row backgrounds for scan-ability, consistent padding, and subtle bottom borders. Pass percentage columnWidths like '30%' and keep column counts low – this book tops out at four before LETTER or A4 gets cramped. The kit ships this recipe as the Table component (src/components/Table.tsx).

```
tsx
const DataTable = ({ headers, rows, columnWidths }) => (
  <View style={styles.tableContainer}>
    /* Header row */
    <View wrap={false} style={styles.tableHeader}>
      {headers.map((h, i) => (
        <Text key={i} style={[styles.tableHeaderText,
          { width: columnWidths[i] }]}>{h}</Text>
      ))}
    </View>
    /* Data rows with alternating colors */
    {rows.map((row, ri) => (
      <View key={ri} wrap={false} style={ri % 2 === 1
        ? styles.tableRowAlt : styles.tableRow}>
        {row.map((cell, ci) => (
          <Text key={ci} style={[styles.tableCell,
            { width: columnWidths[ci] }]}>{cell}</Text>
        ))}
      </View>
    ))}
  </View>
);
```

Every workaround in this chapter follows the same pattern: the CSS property you want doesn't exist, so you build it from View, Text, and flexbox. That's not a limitation – it's the entire design philosophy. Once you stop reaching for web-centric CSS and start thinking in react-pdf primitives, layouts get faster to build, not slower.

CHAPTER 08

Icons over Emojis

SVG icons render sharp, match your brand, and work offline.

Emoji Trade-offs in PDFs

Emojis look harmless. They're quick to type and AI agents love inserting them. But in react-pdf, they introduce real problems:

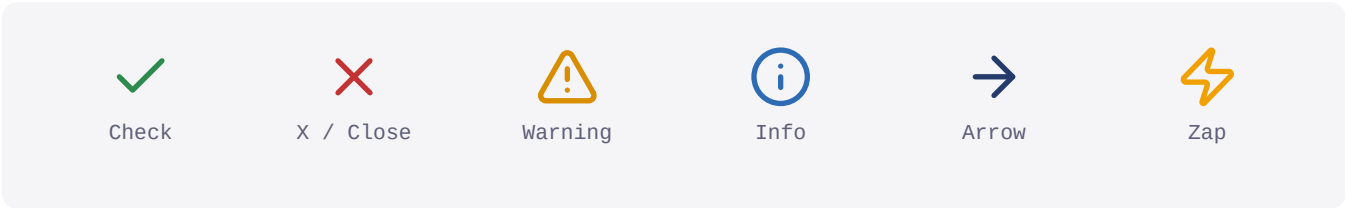
Issue	Impact
Needs an emoji image source	Remote sources add network risk; vendored sources work offline
Raster images at fixed resolution	Blurry when zoomed – PDFs are vector-first
Source-dependent artwork	A fixed source renders consistently across platforms
Color baked into the artwork	Cannot be recolored with document tokens
Informal appearance	Undercuts professional documents
Remote CDN outage	Breaks builds only when the selected source is remote

⚠️ Remote Source Risk

Font.registerEmojiSource() can use a remote set such as Twemoji or a vendored local image set. A remote source makes builds depend on the network and its host; vendor the assets when offline and reproducible rendering matter.

SVG Icon Advantage

React-PDF has extensive SVG support. SVG icons are vector graphics – they scale cleanly at any zoom level, match any color, and render consistently across PDF viewers with no network or CDN dependency at render time.



These icons come from Lucide via the react-icons package (ISC license). A small adapter rewrites react-icons' browser SVG output as @react-pdf/renderer Svg/Path nodes. Size and color stay under your control; stroke weight follows the source icon unless you expose it as another adapter prop.

Why Not Emojis?

Property	SVG icon	Emoji glyph
Color	Any token you want	Fixed by the artwork
Rendering	Consistent vector path	Depends on chosen source
Stroke weight	Set by source icon	None
Print fidelity	Crisp vectors	Often missing

react-icons Adapter

react-icons returns plain HTML `<svg>/<path>` elements, which react-pdf can't render. The adapter walks the icon's React tree once and rebuilds it with react-pdf primitives, normalizing `currentColor` and string-typed numerics along the way.

```
tsx
import { Svg, Path, Circle, Line } from '@react-pdf/renderer';
import type { IconType } from 'react-icons';
import { iconSize } from '../styles/theme';
const TAG_MAP = { svg: Svg, path: Path, circle: Circle, line: Line /* ...etc */ };

const Icon = ({ icon, size = iconSize.lg, color }) => {
  const { attr, children } = icon({}).props;
  const svgProps = coerceProps(attr, color);
  return (
    <Svg width={size} height={size} {...svgProps}>
      {convertChildren(children, color, svgProps)}
    </Svg>
  );
};
```

Using Icons in Context

- ✓ Vector-sharp at any zoom level
- ✓ Matches your brand color palette
- ✓ Zero network dependencies at render time – no CDN, no internet

⚡ When Emojis Are Acceptable

Internal team documents, quick prototypes, or documents where your brand literally uses emoji as part of its identity. For anything client-facing or paid – use SVG icons.

Skip the Path Data – Use react-icons

The book's source ships an Icon adapter for compatible react-icons IconType values built from its mapped SVG primitives. The package offers nearly 50,000 vetted icons (Lucide, Heroicons, Feather, Tabler, Font Awesome) – no path-string copying or SVG hand-tuning for compatible icons.

```
tsx
import { LuCheck, LuTriangleAlert } from 'react-icons/lu';
import Icon from './components/Icon';
import { colors, iconSize } from './styles/theme';
<Icon icon={LuCheck} size={iconSize.md} color={colors.success} />
<Icon icon={LuTriangleAlert} size={iconSize.lg} color={colors.warning} />
```

Re-export the icons your document actually uses from a single Icons.tsx so prompts stay short. Keep that re-export file under ~12 icons; ad-hoc **react-icons/lu** imports are fine for one-off uses.

⚠ Path Data Quality

Not every icon library renders cleanly. Stick to stroke-based sets (Lucide, Feather, Heroicons-outline). Avoid icons that rely on filters or masks – react-pdf supports neither, and the adapter skips any tag it can't map, logging a one-time build warning that names it.

Which Sets Render Cleanly

Import prefix	Library	Style	Use it?
react-icons/lu	Lucide	Stroke outline	Yes – first choice
react-icons/fi	Feather	Stroke outline	Yes – minimal set
react-icons/hi2	Heroicons v2	Outline + solid	Yes – use outline

Sizing Cheat Sheet

Match icons to the text beside them – oversized looks amateur, undersized disappears.

Context	Icon Size	Text Size
Inline with body text	12-14pt	11pt body
List markers	6pt dot / 10-12pt icon	9.5-11pt
Section headers	16pt	20pt h2
Feature showcases	24pt	Display or standalone
Callout box labels	13pt	13pt label

Emoji vs Icon, Side by Side

Property	Emoji	SVG Icon
Rendering	Source-dependent	Consistent vector path
Sizing	Fixed to font size	Any size, precise control
Color	Cannot be changed	Any color from your palette
Print quality	May rasterize poorly	Vector – infinite resolution
AI consistency	Depends on source setup	Deterministic output

CHAPTER 09

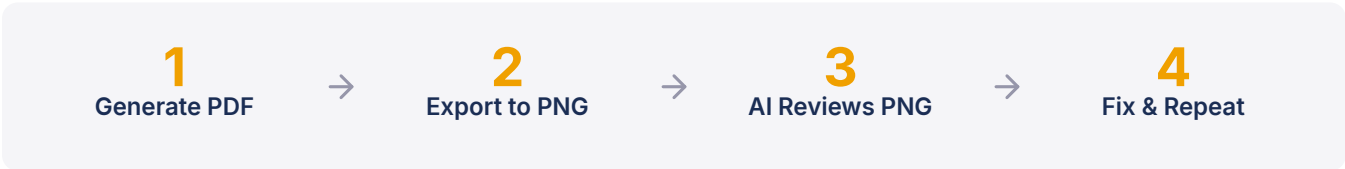
AI Visual Analysis

Export to PNG. Let AI see what the reader sees. Fix what doesn't look right.

Why PNGs Beat Raw PDFs for AI Review

AI tools preprocess PDFs differently, often combining extracted text with page images. For visual QA, control that conversion: export PNGs at a chosen DPI so the model reviews a consistent rasterization of the same PDF layout your readers receive.

Visual QA Workflow



PDF to PNG Conversion

The most reliable tool is pdftoppm from the poppler-utils package. It's fast, handles edge cases well, and produces clean output. DPI trades image quality against file size – 200 DPI is the sweet spot. Sizes below are measured from this book's text-heavy pages.

DPI	Use Case	File Size
72	Quick preview, thumbnail	~15-105KB per page
150	Standard review, good detail	~35-265KB per page
200	Detailed AI analysis (recommended)	~50-320KB per page
300	Print-quality verification	~80-440KB per page

Export Script

Wire the export into package.json: "pnpm build" generates the PDF, "pnpm export" rasterizes it to PNGs, and "pnpm pipeline" chains both. The script below is simplified – the full version also checks for a missing PDF or pdftoppm.

```
bash
#!/bin/bash
# scripts/export-pages.sh (simplified)
set -e
PDF_FILE="output/react-pdf-ai-builders-guide.pdf"
PNG_DIR="output/pages"
DPI="${1:-200}" # DPI argument, default 200
mkdir -p "$PNG_DIR"
rm -f "$PNG_DIR"/page-*.png # clear stale pages
pdftoppm -png -r "$DPI" "$PDF_FILE" "$PNG_DIR/page"
echo "Exported to $PNG_DIR/"
ls -lh "$PNG_DIR"/page-*.png
```

What AI Can Spot

Be specific in your prompts: "Check if the heading alignment is consistent across pages 3-7" works better than "review this PDF." Review in small batches – assistants enforce image and request-size limits, so open only the pages you changed. Here's what vision models reliably detect:

- ✓ Alignment inconsistencies between elements
- ✓ Spacing irregularities (uneven margins, padding gaps)
- ✓ Font size mismatches across similar elements
- ✓ Color contrast issues (text hard to read on background)
- ✓ Layout balance problems (too heavy on one side)
- ✓ Missing or broken elements
- ✓ Orphaned lines at page breaks
- ✓ Overall visual hierarchy assessment

What AI Struggles With

- ✗ Exact pixel measurements (estimates within ~5-10%)
- ✗ Precise color matching (close, not pixel-perfect)
- ✗ Specific font identification
- ✗ Reading small text at low DPI
- ✗ Distinguishing very similar shades

Here's what matters: AI catches relative problems, not absolute ones. It'll tell you "this heading looks different from that heading" but not "this margin is exactly 24px." Use it for what it's good at – alignment, spacing rhythm, visual hierarchy, broken elements – and verify exact values in your code.

Three Review Prompts That Work

Chapter 6 covered prompts that **generate** pages. These are different – they ask AI to **review** a page that already exists. Single-axis prompts get you specifics:

`text`

`SPACING: "Look at this PDF page. Are the vertical gaps between sections consistent? Are any two adjacent elements too close together or too far apart?"`

`HIERARCHY: "This page should have 4 levels of text: page title (largest), section headings, body text, and captions (smallest). Can you clearly identify all 4 levels? Are any two levels too similar in size?"`

`CONSISTENCY: "Here are pages 3 and 4 of the same document. Do the headers look identical? Are the margins the same? Do the callout boxes use the same styling? Flag any inconsistencies between the pages."`

CHAPTER 10

Premium Deliverables & Recipes

The checklist that separates a free PDF from a \$50 one.

Premium Checklist

Every item below is concrete and verifiable. A PDF that hits all of these looks intentionally designed – not generated.

Typography

- ✓ Custom font registered (not default Helvetica/Arial)
- ✓ 3+ distinct text sizes used with clear hierarchy
- ✓ Consistent heading styles across all pages
- ✓ Body text at readable size (10-12pt) with 1.5-1.6 line height
- ✓ Code blocks in monospace font with background color

Color & Visual

- ✓ Palette limited to 3-5 core color roles plus intentional tonal shades
- ✓ Primary color for headings and emphasis areas
- ✓ Accent color for highlights, callout borders, interactive elements
- ✓ Neutral scale for body text, borders, and backgrounds
- ✓ Semantic colors carry meaning; decorative accents stay within the brand palette

Production & Polish

- ✓ Custom fonts embedded; standard PDF base fonts used intentionally
- ✓ File size kept lean – images compressed, no unused assets
- ✓ Selectable, copyable text (not rasterized into images)
- ✓ Final PDF proofed at 100% zoom and at print scale

Layout & Spacing

- ✓ Generous page margins (50-70pt on all sides)
- ✓ Consistent spacing scale used throughout (4pt base with micro-adjustments)
- ✓ Whitespace used intentionally – pages don't feel cramped
- ✓ No orphaned single lines at page breaks
- ✓ Visual breaks between major sections

Structure & Navigation

- ✓ Cover page with strong visual identity
- ✓ Table of contents with chapter listing
- ✓ Page numbers on every content page
- ✓ Headers with section context
- ✓ Footer with branding
- ✓ Chapter title pages with distinct design

Content Components

- ✓ Callout boxes for tips, warnings, and key information
- ✓ Styled code blocks with language labels
- ✓ Professional tables with header rows and alternating colors
- ✓ SVG icons (not emojis) for visual elements
- ✓ Bullet lists with consistent formatting
- ✓ Document metadata set (title, author, subject)

Quality Tests

- The squint test: step back, squint at the page. Can you see clear visual hierarchy? If everything blurs into one gray block, you need more contrast.
- The scroll test: flip through all pages quickly. Do they feel like one cohesive document? Or do some pages look like they belong to a different PDF?
- The "would I share this?" test: if someone you respect saw this PDF with your name on it, would you be proud? If you hesitate, it needs more work.
- The comparison test: open a professionally designed PDF from a publisher. Hold yours next to it. Where does yours fall short? Fix those specific things.

From Checklist to Habit

The first time through this checklist takes effort. By the third document, it becomes muscle memory. You'll register custom fonts automatically, reach for your design tokens without thinking, and notice spacing inconsistencies at a glance. The goal isn't to check boxes forever – it's to internalize what premium looks like so your first drafts start closer to the finish line.

⚡ The Design System Payoff

If you built your theme.ts correctly and pages consume it directly or through shared styles and components, most of this checklist is satisfied automatically. The design system isn't extra work – it's the shortcut that makes every page look premium by default.

Automated AI Review

The manual loop – export PNGs, look at them with AI – automates cleanly. A small script sends each page image to Claude with your quality rubric and prints a score, so every build gets a consistent design pass instead of a hand review you might skip.

```
tsx
import Anthropic from '@anthropic-ai/sdk';
import { readFileSync, readdirSync } from 'fs';

const claude = new Anthropic();
const RUBRIC = 'Score 1-10 on hierarchy, spacing, and consistency. List any issues.';

const pages = readdirSync('output/pages').filter((f) => f.endsWith('.png'));
for (const file of pages) {
  const data = readFileSync(`output/pages/${file}`).toString('base64');
  const res = await claude.messages.create({
    model: 'claude-opus-4-8',
    max_tokens: 1024,
    messages: [{ role: 'user', content: [
      { type: 'image', source: { type: 'base64', media_type: 'image/png', data } },
      { type: 'text', text: RUBRIC },
    ] }],
  });
  console.log(file, res.content[0].type === 'text' && res.content[0].text);
}
```

⚡ Wire it into CI

Run pnpm pipeline in a GitHub Actions job, upload output/ as an artifact, then run this script so every PR gets a design score. Review only changed pages to keep cost and token use low.

Free vs. Premium at a Glance

Signal	Free / Generic	Premium / Polished
Fonts	Default Helvetica	Custom registered typeface
Colors	Random hex values	Systematic palette with intent
Spacing	Inconsistent margins	4pt base with micro tokens
Components	Plain text + emoji	Styled boxes with SVG icons
Hierarchy	Everything same size	Clear heading scale with accents

Recipes & Templates

These recipes turn the checklist into code. They use Chapter 4's tokens and Chapter 3's shared components; the invoice also reuses Chapter 7's Table. Review those chapters or clone the source if you're starting here.

Three recipes follow: an invoice component (the most-requested PDF use case), a data-driven page generator, and a layout patterns cheat sheet. Each is a self-contained pattern you can copy into your project and customize.

Recipe: Invoice Component

This invoice component takes a typed data object and renders a header, line items table, and totals. Swap the InvoiceData interface for any typed shape (receipts, purchase orders, quotes) and the structure stays the same.

```
tsx
interface InvoiceData {
  number: string; due: string;
  from: { name: string; address: string };
  items: { desc: string; qty: number; price: number }[];
  tax?: number;
}

const Invoice = ({ data }: { data: InvoiceData }) => {
  const subtotal = data.items.reduce((s, i) => s + i.qty * i.price, 0);
  const tax = subtotal * (data.tax ?? 0);
  return (
    <Page size="LETTER" style={styles.page}>
      <View style={{ flexDirection: 'row', marginBottom: spacing.xl }}>
        <View style={{ width: '50%' }}>
          <Text style={styles.h3}>{data.from.name}</Text>
          <Text style={styles.bodySmall}>{data.from.address}</Text>
        </View>
        <View style={{ width: '50%', alignItems: 'flex-end' }}>
          <Text style={styles.h2Text}>INVOICE #{data.number}</Text>
          <Text style={styles.bodySmall}>Due: {data.due}</Text>
        </View>
      </View>
      <Table headers={['Description', 'Qty', 'Price', 'Total']}
        rows={data.items.map(i => [i.desc, String(i.qty),
          '$' + i.price.toFixed(2), '$' + (i.qty*i.price).toFixed(2)])}
        columnWidths={['50%', '15%', '15%', '20%']} />
      <View style={{ alignItems: 'flex-end', marginTop: spacing.sm }}>
        <Text style={styles.body}>Subtotal: ${subtotal.toFixed(2)}</Text>
        <Text style={styles.h3}>Total: ${(subtotal+tax).toFixed(2)}</Text>
      </View>
    </Page>
  );
};
```

Recipe: Data-Driven Pages

Most useful PDFs are generated from data, not static content. The pattern: pass a typed array, `.map()` to render one page per item, use conditional rendering to control which sections appear. Each page becomes a pure function of its data – change the input and the PDF rebuilds.

```
tsx
interface Section {
  title: string;
  body: string;
  metrics?: { label: string; value: string }[];
}

const Report = ({ sections }: { sections: Section[] }) => (
  <Document>
    {sections.map((section, i) => (
      <ContentPage key={i} sectionTitle={section.title}>
        <SectionHeading>{section.title}</SectionHeading>
        <Text style={styles.body}>{section.body}</Text>

        {section.metrics && (
          <View style={{ flexDirection: 'row', gap: spacing.md }}>
            {section.metrics.map((m, j) => (
              <View key={j} wrap={false} style={{
                flex: 1,
                backgroundColor: colors.neutral[50],
                padding: spacing.md,
                borderRadius: borders.radius.md,
              }}>
                <Text style={styles.bodySmall}>{m.label}</Text>
                <Text style={styles.h3}>{m.value}</Text>
              </View>
            ))}
          </View>
        )}
      </ContentPage>
    ))}
  </Document>
);
```

Layout Patterns Cheat Sheet

Four layouts you'll reach for repeatedly. Each is a self-contained pattern you can drop into any ContentPage.

Two-Column Layout

```
tsx
<View style={{ flexDirection: 'row', gap: spacing.lg }}>
  <View style={{ flex: 1 }}>
    <Text style={styles.body}>Left column</Text>
  </View>
  <View style={{ flex: 1 }}>
    <Text style={styles.body}>Right column</Text>
  </View>
</View>
```

Sidebar + Content

```
tsx
<View style={{ flexDirection: 'row', gap: spacing.lg }}>
  <View style={{ width: '28%',
    backgroundColor: colors.neutral[50],
    padding: spacing.md,
    borderRadius: borders.radius.md }}>
    <Text style={styles.h4}>Sidebar</Text>
  </View>
  <View style={{ flex: 1 }}>
    <Text style={styles.body}>Main content area</Text>
  </View>
</View>
```


Card Grid (2x2)

```
tsx
<View style={{ flexDirection: 'row', flexWrap: 'wrap',
gap: spacing.md }}>
  {items.map((item, i) => (
    <View key={i} wrap={false} style={{
      width: '48%', padding: spacing.md,
      borderWidth: borders.medium,
      borderColor: colors.neutral[200],
      borderRadius: borders.radius.md,
    }}>
      <Text style={styles.h4}>{item.title}</Text>
      <Text style={styles.bodySmall}>{item.desc}</Text>
    </View>
  ))}
</View>
```

Metric / KPI Row

```
tsx
<View style={{ flexDirection: 'row', gap: spacing.md }}>
  {metrics.map((m, i) => (
    <View key={i} style={{ flex: 1, alignItems: 'center',
padding: spacing.md,
backgroundColor: colors.neutral[50],
borderRadius: borders.radius.md }}>
      <Text style={styles.bodySmall}>{m.label}</Text>
      <Text style={styles.h2Text}>{m.value}</Text>
    </View>
  ))}
</View>
```

Putting It All Together

- Use Chapter 3's file-per-page structure to keep recipes composable
- Use Chapter 4's tokens for reusable colors and spacing in recipe code
- Apply Chapter 2's `wrap={false}` patterns to keep recipe cards from splitting across pages
- Clone the source – every pattern here follows conventions used throughout the book
- Explore the react-pdf GitHub repo for additional component examples and API updates

The invoice and report recipes follow typed data in, styled PDF out: interfaces define the contract, components handle layout, and tokens enforce consistency. The layout recipes instead demonstrate reusable composition without a data model. Start with the pattern closest to your use case and build from there.

The Shape of a Data-Driven Recipe

Strip away the specifics and each data-driven recipe reduces to the same three lines. Hand your AI this skeleton and the recipe page it should follow – it will fill in the interface and layout against your real data.

```
tsx
interface ReportData { /* your typed contract */ }

const Report = ({ data }: { data: ReportData }) =>
<Document><Page style={styles.page}>{/* tokens + components */}</Page></Document>;
```

⚡ One Pattern, Many Documents

Invoices, reports, certificates, dashboards – they all share this skeleton. Once your AI internalizes the typed-data-in, styled-PDF-out shape, generating a new document type is mostly a matter of swapping the interface and picking which components to compose.

CHAPTER 11

Troubleshooting & Common Errors

The errors you'll hit, why they happen, and how to fix them fast.

Raw Strings Outside a Text Component

⚠ Console Warning

"Invalid '...' string child outside <Text> component"

This is the most common react-pdf mistake. In this version, a raw string directly under Page or View triggers a warning and is omitted. Put ordinary copy in Text; text-capable primitives such as Link are exceptions.

```
tsx
// Broken - raw string inside View
<View>
This raw string is invalid
</View>

// Fixed - wrap in Text
<View>
<Text style={styles.body}>This renders correctly</Text>
</View>
```

Fonts Not Loading

⚠ Symptom

The build reports an unregistered family or unreadable font file, or text uses a different weight than expected.

An unknown family or bad file path fails the build. For an unavailable weight, react-pdf selects the nearest registered weight. Helvetica is the default only when neither Text nor an ancestor sets fontFamily; this project's ContentPage inherits Inter from its Page style.

Content Overflows the Page

⚠ Symptom

Content runs off the bottom of the page, disappearing below the margin. Footer may overlap with body text.

This project renders one non-wrapping ContentPage per source file. If one overflows, shorten or split that file. In a wrapping document, a wrap=false parent or fixed content height can block page breaks.

```
tsx
// On a wrapping Page, this child cannot break
<View wrap={false} style={{ minHeight: 800 }}>
  /* Content can't split - overflows the page */
</View>
// Fix there: allow the parent to wrap, lock each item
<View> /* wrap={true} is default */
  {items.map((item, i) => (
    <View key={i} wrap={false}> /* Each item stays together */
    <Text style={styles.body}>{item}</Text>
    </View>
  ))}
</View>
```

Other Culprits to Check

Property	Why it overflows	Fix
height / minHeight	Forces a content box taller than the page can hold.	Remove it from content containers.
position: absolute	Escapes flow, so react-pdf cannot break it.	Use flex layout for body content.

Orphaned Headings

Symptom

A section heading appears at the very bottom of a page with no content following it. The body text starts on the next page.

On a wrapping Page, minPresenceAhead moves the heading forward when too little content can follow it. This project's ContentPage is non-wrapping, so prevent orphans by shortening or splitting the source page during visual QA.

```
tsx
{/* Use inside a wrapping Page */}
<View wrap={false} minPresenceAhead={40}>
<Text style={styles.h2Text}>Section Title</Text>
</View>
```

Split Callout Boxes

Symptom

A colored callout box splits across a page break. The top half shows the colored background, the bottom appears without – looking broken.

Callout boxes should always stay on a single page. The TipBox, WarningBox, and InfoBox components include wrap=false internally. For custom callouts, add wrap=false to the outer View.

Styles Not Applying

react-pdf supports a subset of CSS. Property names must be camelCase, and several common web CSS properties are not available:

Web CSS	react-pdf Equivalent
box-shadow	Not supported – use borderWidth + borderColor
text-decoration	textDecoration (underline, line-through)
linear-gradient	Not supported – use Svg + Defs
display: grid	Not supported – use flexbox
font-weight: bold	fontWeight: 700 (numeric or named)

Flexbox Layout Surprises

⚠ Symptom

Elements stack vertically when you expect them side-by-side. Default flexDirection is "column", not "row".

Default: column

width: 50%

width: 50%

Fixed: row

50%

50%

Build and Memory Errors

⚠ Symptom

Build crashes with "JavaScript heap out of memory" or takes extremely long to complete.

Large documents with many images consume significant memory. Node's default heap (typically 2–4 GB, sized from system memory) may not be enough for image-heavy PDFs past ~50 pages.

```
bash
# pnpm build still runs pnpm sync first
NODE_OPTIONS=--max-old-space-size=4096 pnpm build
```

- Resize images to actual display size before embedding (2x for retina)
- Use JPEG for photos (quality 80), PNG only for graphics with transparency
- For 50+ page documents, consider splitting into multiple PDF files

Slow Builds: Find the Culprit

Symptom	Likely cause	Fix
Build hangs early	Remote image URLs blocking on the network	Download assets once, embed locally
Steady RAM climb	Full-resolution images embedded as-is	Pre-resize to display size, prefer JPEG
Slow on every run	Re-registering fonts per render	Register fonts once at module load

Debug Workflow

When something looks wrong in your PDF, follow this sequence:

- Simplify: comment out everything except the broken section. Does it still break in isolation?
- Isolate: move the broken component into a standalone single-page document. Render it alone.
- Check wrap: add `wrap={false}` to containers that should stay together, remove it from containers that need to split.
- Check fonts: temporarily switch to `fontFamily: "Helvetica"` to rule out font issues.
- Check styles: log style objects to console. Are values what you expect? Are you using camelCase?
- Rebuild incrementally: add sections back one at a time until the break reappears. That's your culprit.

⚡ The Fastest Fix

Most react-pdf layout issues come from three sources: missing `wrap=false` on elements that should stay together, missing Text wrappers around strings, and `flexDirection` defaulting to column. Check these three things first before diving deeper.

Decode the Error

Error message	What it usually means
Invalid string child	A raw string sits outside a Text – wrap it.
fontFamily ... not registered	The family was never registered – import fonts.ts before render.
Cannot read ... of undefined	A style token resolved to undefined – check the import path.

Closing the Loop

Chapter 1 started with a simple premise: professional-grade PDF generation is achievable with AI if you apply the right constraints. Every error above has a deterministic solution, and every solution reinforces the architectural patterns we've covered:

Error	Root Fix	Pattern From
Text not in <Text>	Wrap all strings	Ch2: Fundamentals
Wrong font	Check family/path; register intended weights	Ch2: Fundamentals
Content overflow	Shorten/split the fixed source page	Ch2: Fundamentals
Orphaned heading	Edit/split the fixed source page	Ch2: Fundamentals
Split callout	wrap={false} on outer View	Ch2: Fundamentals
Styles ignored	Use camelCase, check support	Ch2: Fundamentals
Wrong layout	Set flexDirection: "row"	Ch7: Challenges
Out of memory	Resize images, increase heap	Ch11: Troubleshooting

The debug workflow isn't something you do when things go wrong. It's the last step in the build loop: write, render, export, review, fix. When that loop is fast – and with react-pdf and PNG export, it's very fast – you stop fearing bugs and start shipping.

CHAPTER 12

Markdown Automation

Markdown authoring with shared components and explicit page breaks.

The Power of Markdown

Writing content in React components is powerful but tedious. With a Markdown-to-JSX pipeline, you can focus on writing while shared components handle styling; explicit markers still control physical page breaks.

Why Markdown?

- **Focus on content** – no more JSX syntax errors while writing prose.
- **Portable** – your content lives in standard .md files.
- **AI Friendly** – AI models are excellent at writing and editing Markdown.

How it Works

The `MarkdownRenderer` component parses your Markdown and maps it to the project's premium component library.

```
tsx
import { MarkdownRenderer } from '../components';
import fs from 'fs';
const content = fs.readFileSync('content/chapters/12-markdown-demo.md', 'utf-8');
const body = content.replace(/^---[\s\S]*?---/, '').trim();
const pages = body.split('\n<!-- page-break -->\n');
const page = (index: 0 | 1) => (
  <ContentPage sectionTitle="Automation" wrap={false}>
    <MarkdownRenderer content={pages[index].trim()} />
  </ContentPage>
);
```

⚡ Pro Tip

You can mix and match Markdown with custom JSX for complex layouts that Markdown can't handle.

Supported Elements

The current parser supports:

- **Headings** (h1, h2, and h3)
- **Body text** with automatic paragraph grouping
- **Bullet lists** using the `BulletList` component
- **Code blocks** with syntax highlighting and a language label
- **Callouts** (Tip, Warning, Info boxes)

⚠ Keep code blocks short

Fenced blocks render through `CodeBlock`, which uses `wrap={false}` – a block that outgrows the page cannot split, so size each sample to the remaining page height.

Inline Formatting

You can mix **bold**, *italic*, and `inline code` directly in any paragraph, and the renderer maps each run to the matching text style. Reach for *emphasis* where a word carries weight, and `code` whenever you name a prop, file, or value.

📘 Round-trip ready

Because the source is plain Markdown, your AI agent can draft, edit, and review chapters as text – then this pipeline renders them into the same premium components every hand-built page uses.

Now Ship It.

Twelve chapters, from architecture to automation. You have the patterns, the source code, and the templates. The distance between a basic AI-generated PDF and a high-end, professional deliverable is shorter than you think.

- 1** Structure your project for AI – one file per page, design tokens in one place, components that compose (Ch 3).
- 2** Define your design language once – colors, typography, spacing, borders. The theme file enforces consistency so you don't have to (Ch 4).
- 3** Optimize for token budgets – small, focused files mean small, focused prompts that keep AI in its high-attention zone (Ch 5).
- 4** Export to PNG for visual QA – controlled rasterization gives AI a consistent view of reader-visible layout. The loop is generate, export, review, fix (Ch 9).
- 5** Iterate past the first draft – premium output takes 2-3 passes. Use the recipes and checklists from Chapter 10 to close the gap.
- 6** Troubleshoot methodically – check fixed-page overflow, missing Text wrappers, unsupported styles, and column-default flexbox (Ch 11).
- 7** Automate your workflow – use Markdown for content, shared components for styling, and explicit markers for page breaks (Ch 12).

The source code for this entire book is included:

- A design token system (theme.ts) ready to customize.
- A reusable component library – ContentPage, CodeBlock, Table, TipBox, Icons.
- A build and PNG export pipeline.
- AI-optimized reference documentation.

Clone it. Change the colors. **Build your own.**